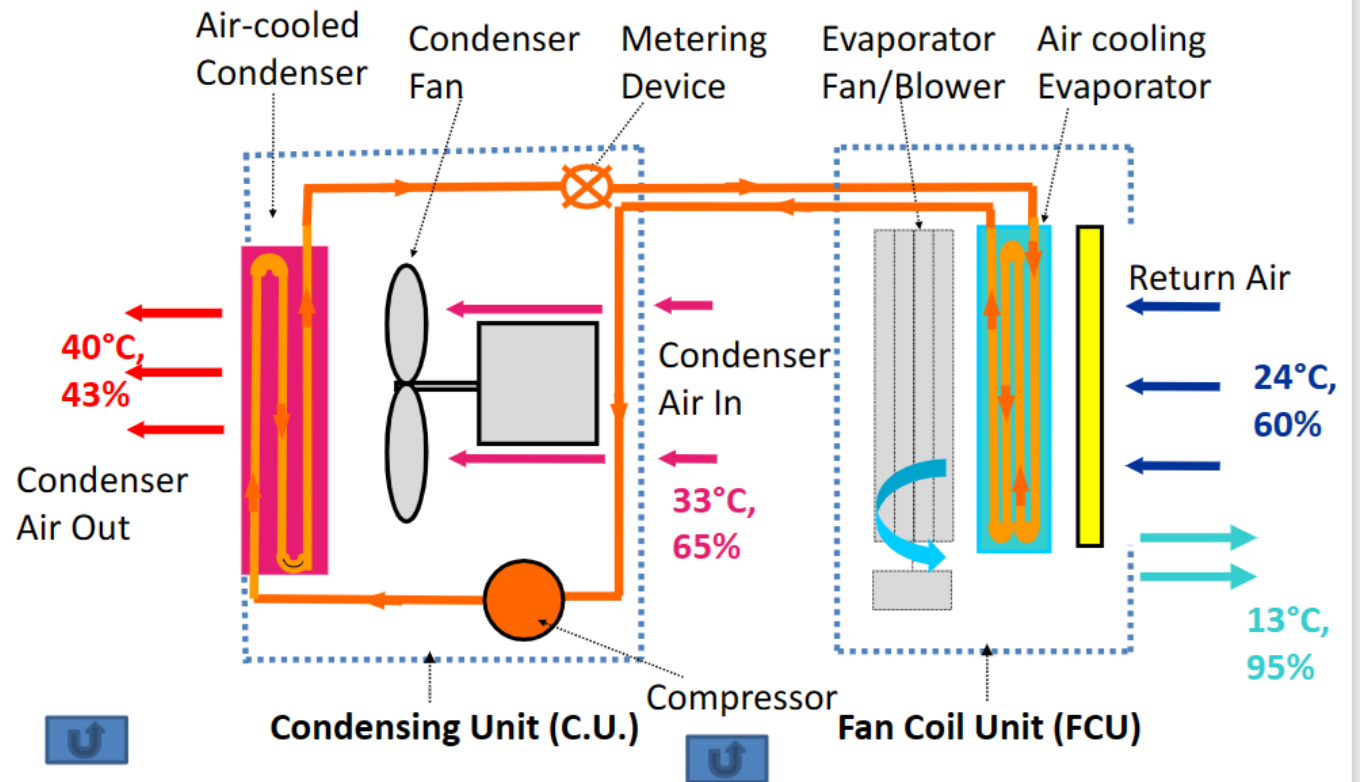


ACMV System description

Air-cooled Split Unit



Objectives

After studying this topic, you shall be able to:

- Explain the process of evaporation and condensation of refrigerants
- Understand the pressure enthalpy diagram of R134A
- Understand refrigeration cycle
- Calculate refrigeration cycle performance

Refrigeration Process

The four refrigeration processes:

1. EVAPORATION
2. COMPRESSION
3. CONDENSATION
4. EXPANSION

Evaporation

- evaporation of refrigerant from liquid to vapour absorbs large amount of heat
- evaporation takes place at low temperature and at low pressure
- low pressure is generated by suction effect of the compressor

Compression

- compressor maintains low pressure at the evaporator for refrigerant to evaporate at low temperature to provide cooling
- compressor maintains high pressure at the condenser for refrigerant to condense at high temperature so that it can be easily cooled by ambient air or water

- Condensation

- condensation of refrigerant from vapour to liquid discharge large amount of heat
- condensation takes place at high temperature and at high pressure
- high pressure is generated by the compressor

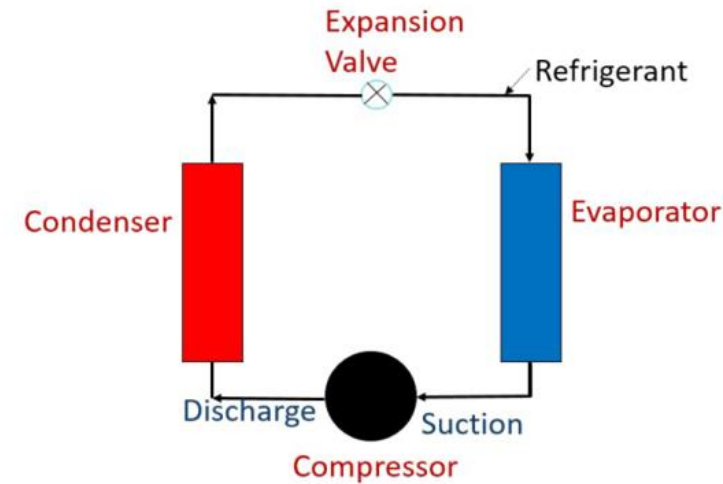
- Expansion

- refrigerant expands from high pressure liquid to low pressure vapour

- Refrigeration System

- 3.2.1. Schematic Diagram

-



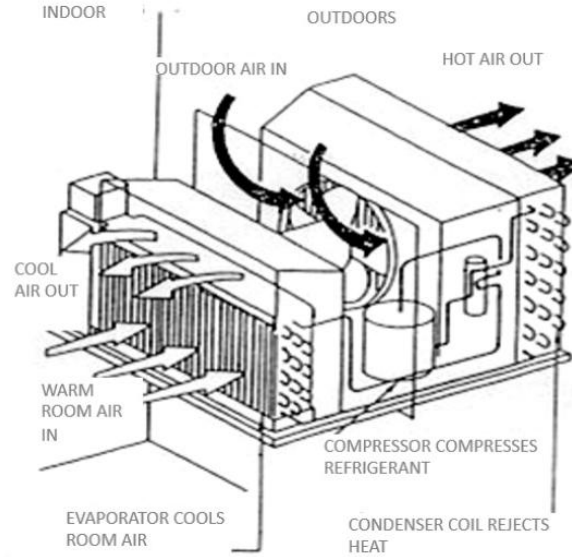
3.2.2. Function of Components

Evaporator – refrigerant evaporates from liquid to vapour and absorbs large amount of heat

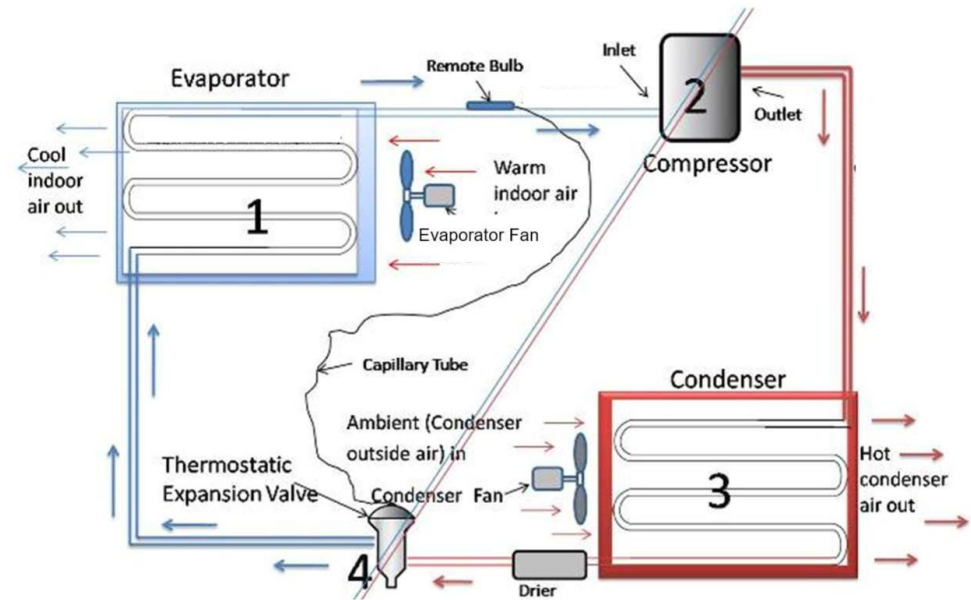
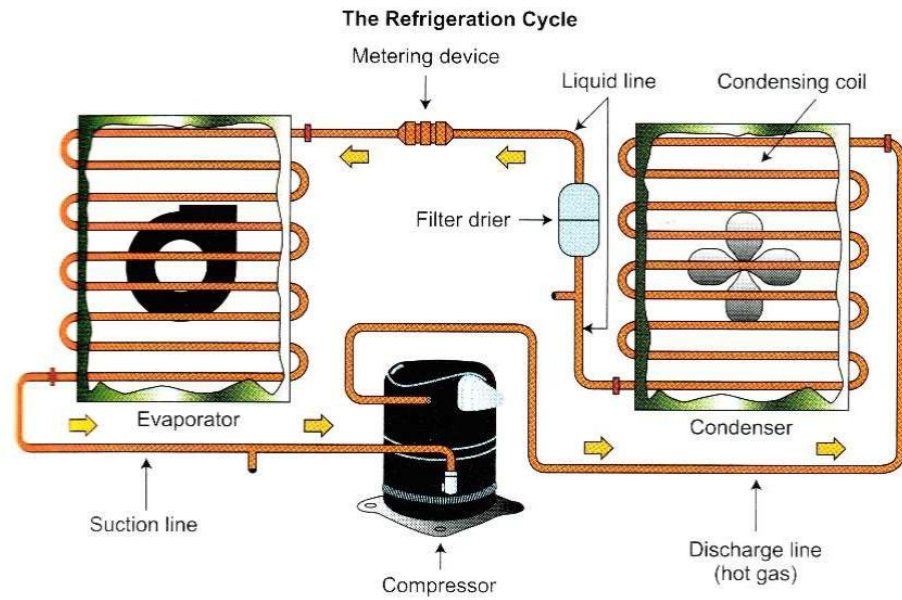
Compressor – refrigerant is compressed from low pressure and low temperature vapour to high pressure and high temperature vapour

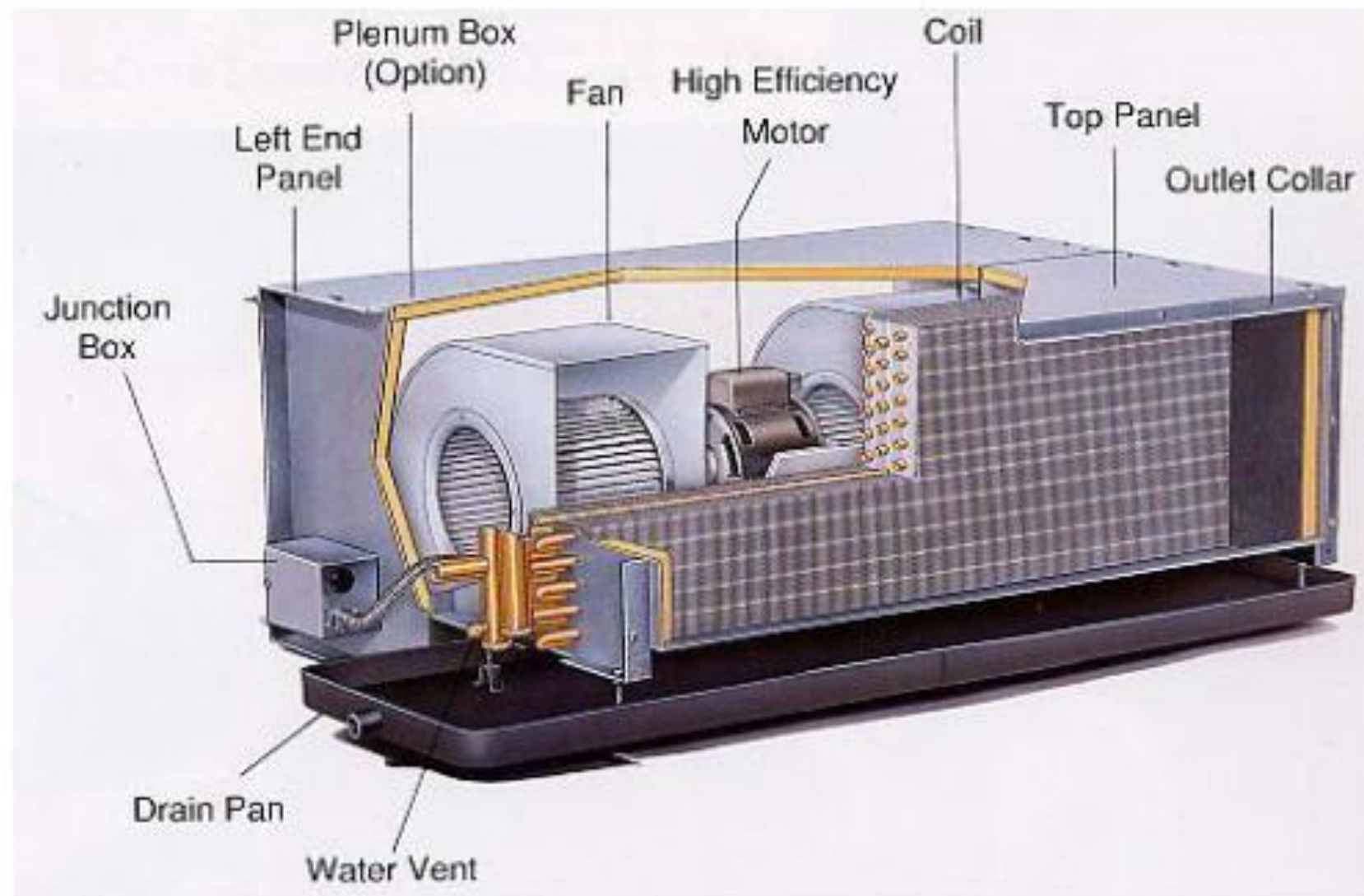
Condenser – refrigerant condenses from vapour to liquid and gives out large amount of heat

Expansion valve (metering device) – when liquid refrigerant passes through the expansion valve it changes from high pressure to low pressure due to the suction effect of the compressor



Window unit (room air conditioner)





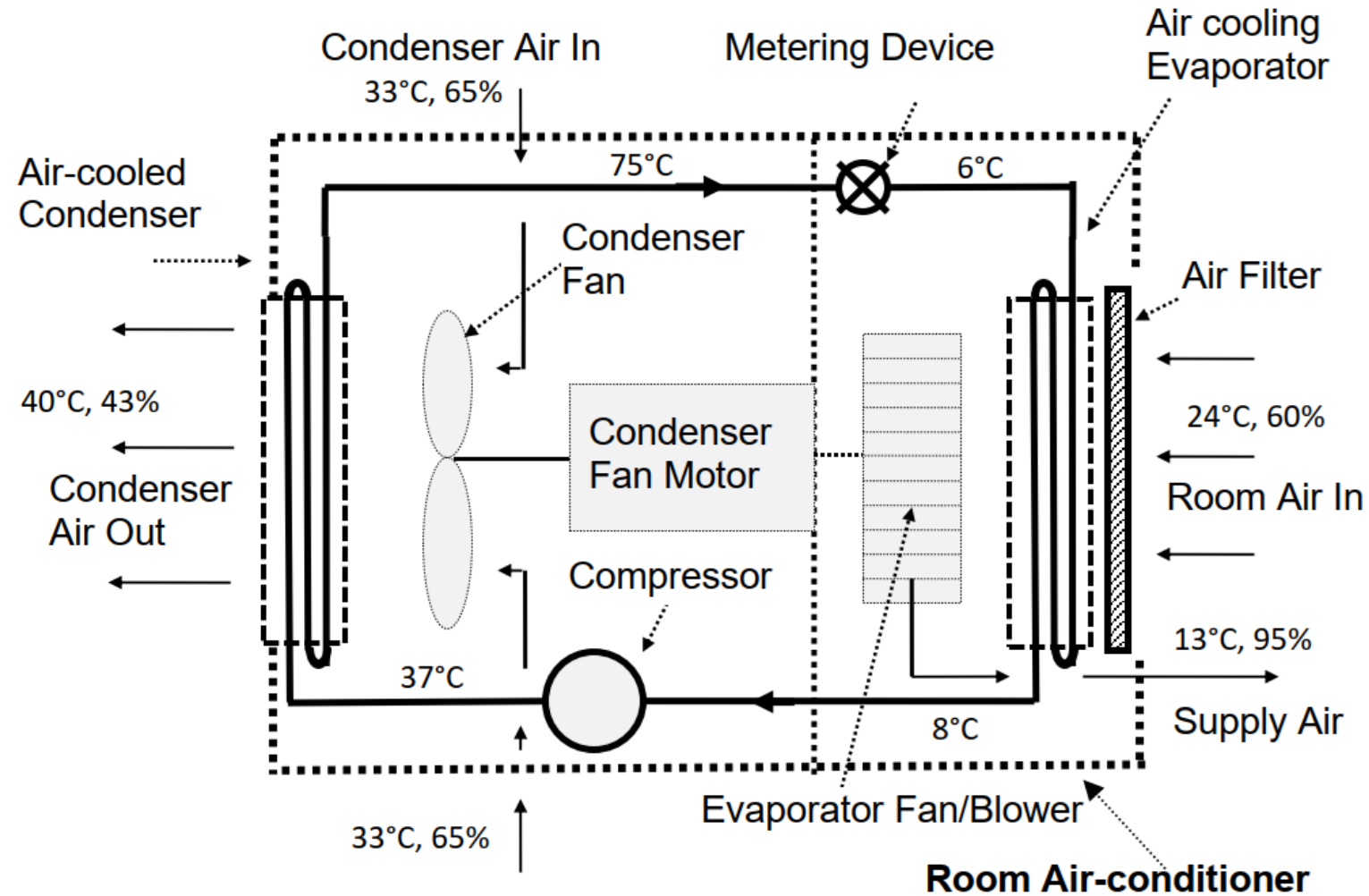
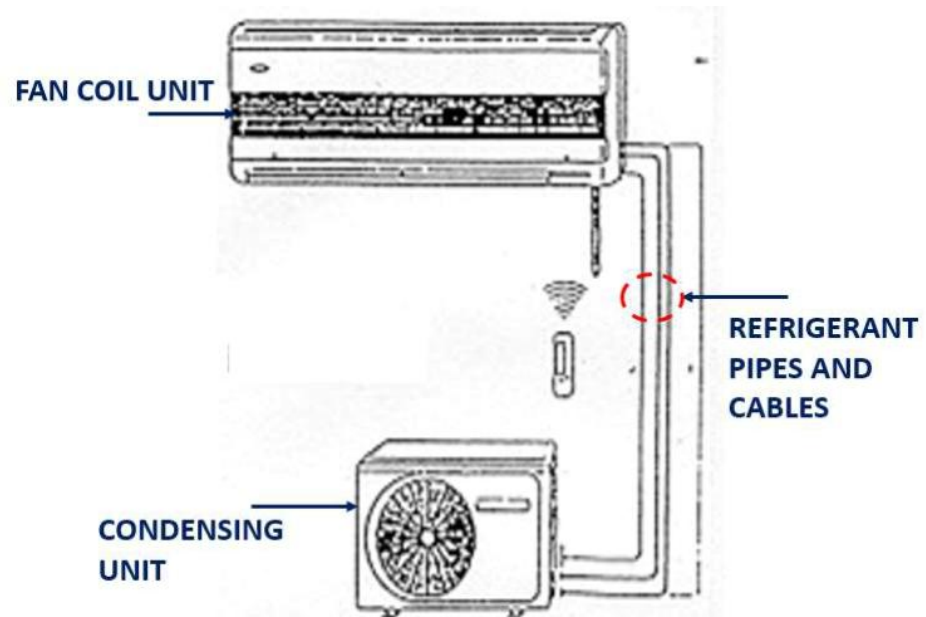
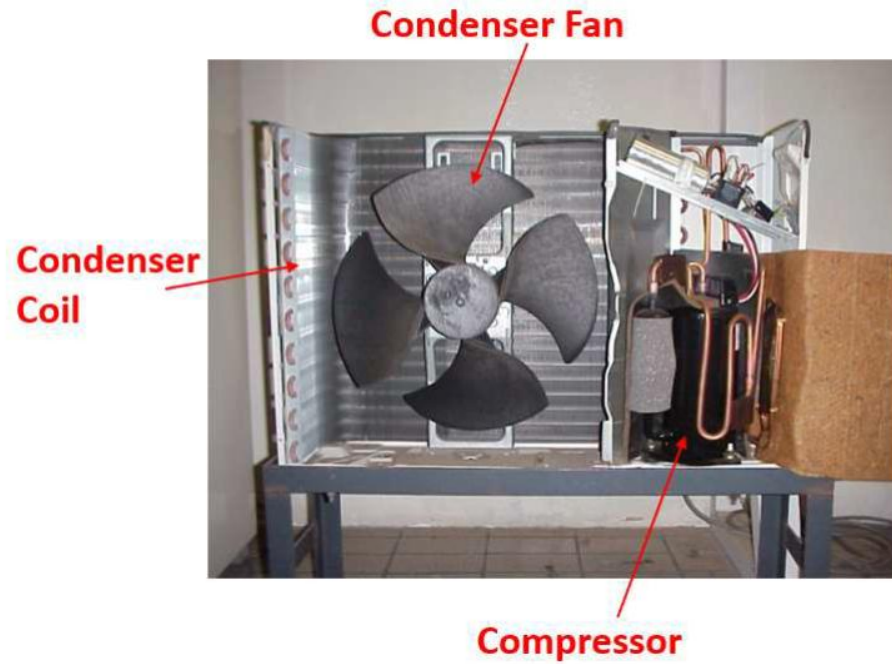


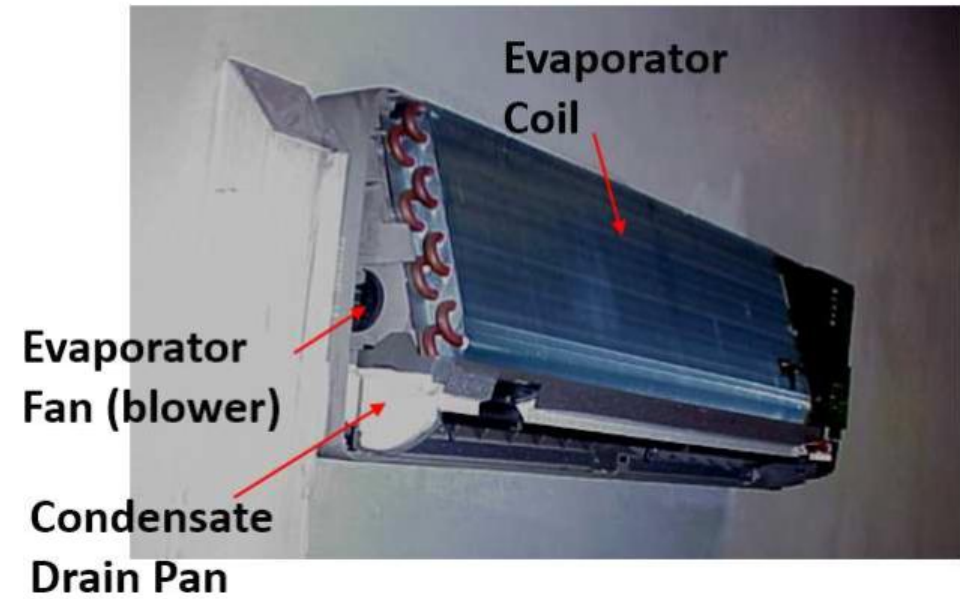
Figure 1.5 – Room Air-conditioner Schematic



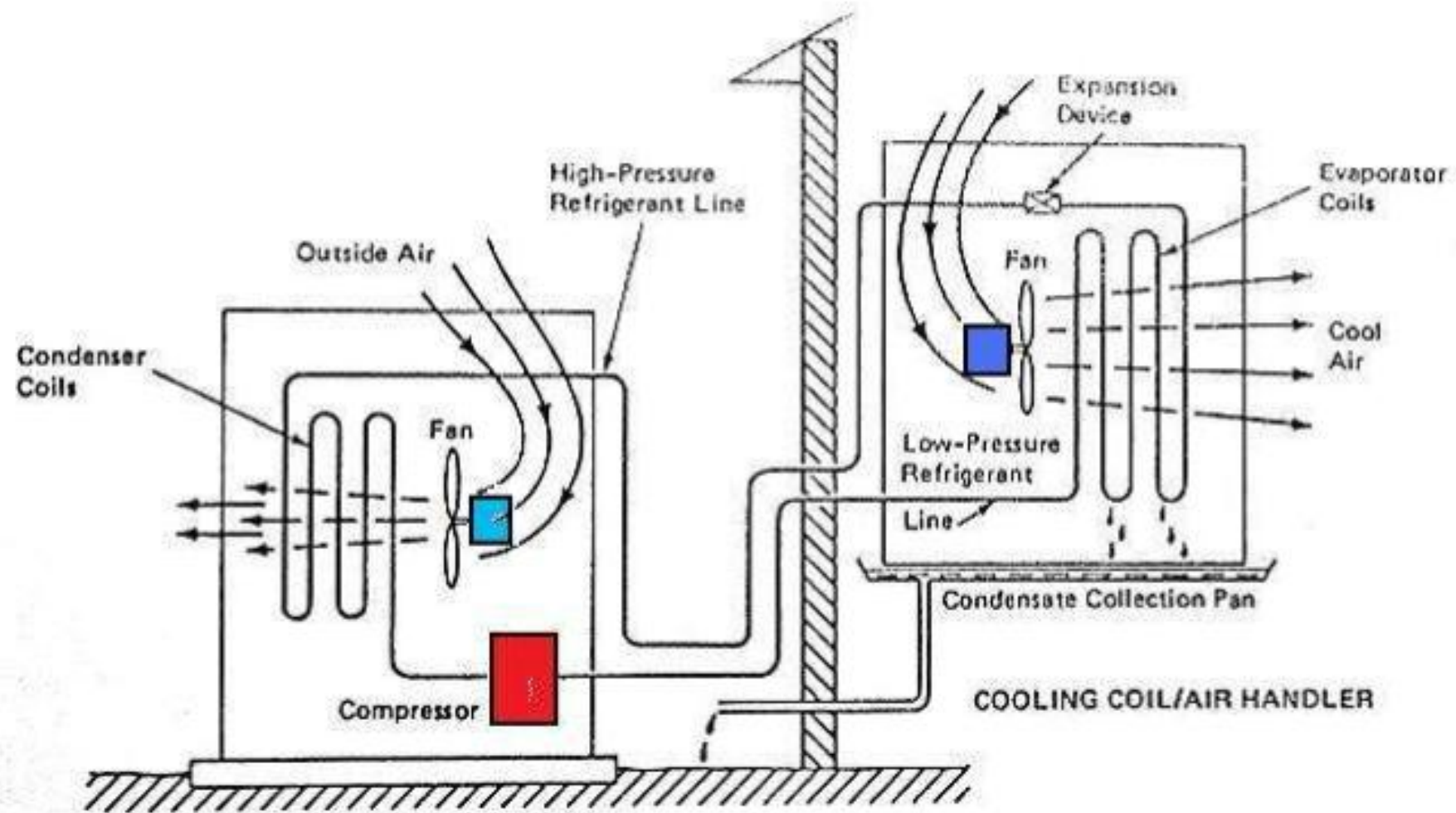
Condensing Unit



Fan Coil Unit







Condensing Units
- compressor,
condenser coil,
condenser fan



Preventive Maintenance

1. Clean filter and front panel.
2. Clean condensate pan and drain pipe.
3. Check electrical controls function.
4. Check running current and on and off-coil air temperatures.
5. Clean evaporator and condenser coils.
6. Clean fan blades.
7. Straighten/comb fins where necessary.
8. Check Gas pressure

Chiller Plant Performance

Performance (efficiency) of chilled water plant is measure by kW/RT.

1 Refrigeration Ton (RT) = 3.517 kW (refrigeration)

The data of chiller plant in Example 1 is shown in the following table.

Chilled Water Return (Inlet) Temperature, t_i (°C) 12.5

Chilled Water Supply (Outlet) Temperature, t_o (°C) 6.7

Chilled Water Mass Flow Rate (kg/s) 2.6

Condenser Water Supply (Inlet) Temperature, t_i (°C) 31.5

Condenser Water Return (Outlet) Temperature, t_o (°C) 38.3

Condenser Water Mass Flow Rate (kg/s) 3.6

Electric Power to Chiller (kW) 32.75

Calculate:

- (i) Chiller cooling capacity
- (ii) Chiller heat rejection
- (iii) Chiller performance (efficiency) in kW/RT

Specific Heat Capacity of Water = 4.19 kJ/kg

1 RT = 3.517 kW

$Q = m \cdot C_p \cdot \Delta T$

1. Chiller cooling capacity = $2.6 \cdot 4.19 \cdot (12.5 - 6.7)$

= 63.185 kW

= 17.966 RT

2. Chiller heat rejection = $3.6 \cdot 4.19 \cdot (38.3 - 31.5)$

= 102.57 kW

3. Electric power input = 32.75 kW

4. Chiller Performance kW/RT = $32.75 / 17.966$

= 1.8229 kW/RT

<https://youtu.be/rgk7UdEWMpQ>